

CLAP-CONTROLLED LED CIRCUIT WITH PROXIMITY DETECTION

PROXIMITY DETECTION

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D h a r w a d

Declaration

I hereby declare that the project entitled “Development of a Double-Layer PCB Based **Clap-Controlled LED Circuit with Proximity Detection**” is my work carried out as part of the academic project requirements. The circuit was designed, simulated/verified where applicable, and developed as a PCB design using electronic design automation tools. The information and references used in this report have been appropriately acknowledged.

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Roll No : 25BEC012

Abstract

The project presents the Development of a PCB-based clap-controlled LED circuit with an additional proximity detection feature. The main objective of the project is to develop a compact electronic circuit in which a clap sound can be detected and used to control an LED output.

The clap signal is detected using a microphone-based sensing stage and processed using transistor circuitry. The resulting pulse is provided to a CD4017 decade counter, which changes its output state according to the detected clap. A transistor switching stage is used to control the LED.

As an additional feature, a proximity detection section is incorporated into the circuit. When a person approaches the system to approximately 1 meter, the proximity sensing section activates the corresponding LED indication. This additional feature improves the practical usability of the system and demonstrates how multiple sensing functions can be incorporated into a single PCB.

The complete circuit was designed using electronic design automation software, beginning with schematic capture and component assignment, followed by PCB layout, routing, design-rule checking, and generation of manufacturing files such as Gerber files. A double-layer PCB is used to satisfy the project requirement and to provide better routing flexibility and compactness.

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Introduction

Conventional electronic switching systems generally require a user to operate a physical switch. Sensor-based control systems provide an alternative approach in which an electrical device can be controlled based on environmental conditions or user actions without direct physical contact.

A clap-controlled circuit is a simple example of sound-based electronic switching. A microphone detects the sound produced by a clap and converts the acoustic signal into an electrical signal. This signal can then be processed and used to control an electronic output.

In this project, a microphone is used as the sound sensing element. A BC547 transistor provides signal amplification and switching, while a 4017 decade counter is used to process the detected clock pulses. Diodes and a second BC547 transistor are used in the output switching section to control an LED.

An additional proximity detection feature has been incorporated into the design. A three-terminal proximity sensor/module provides a DATA output that is combined with the control circuitry. This allows the LED to indicate the presence of a person approaching the system.

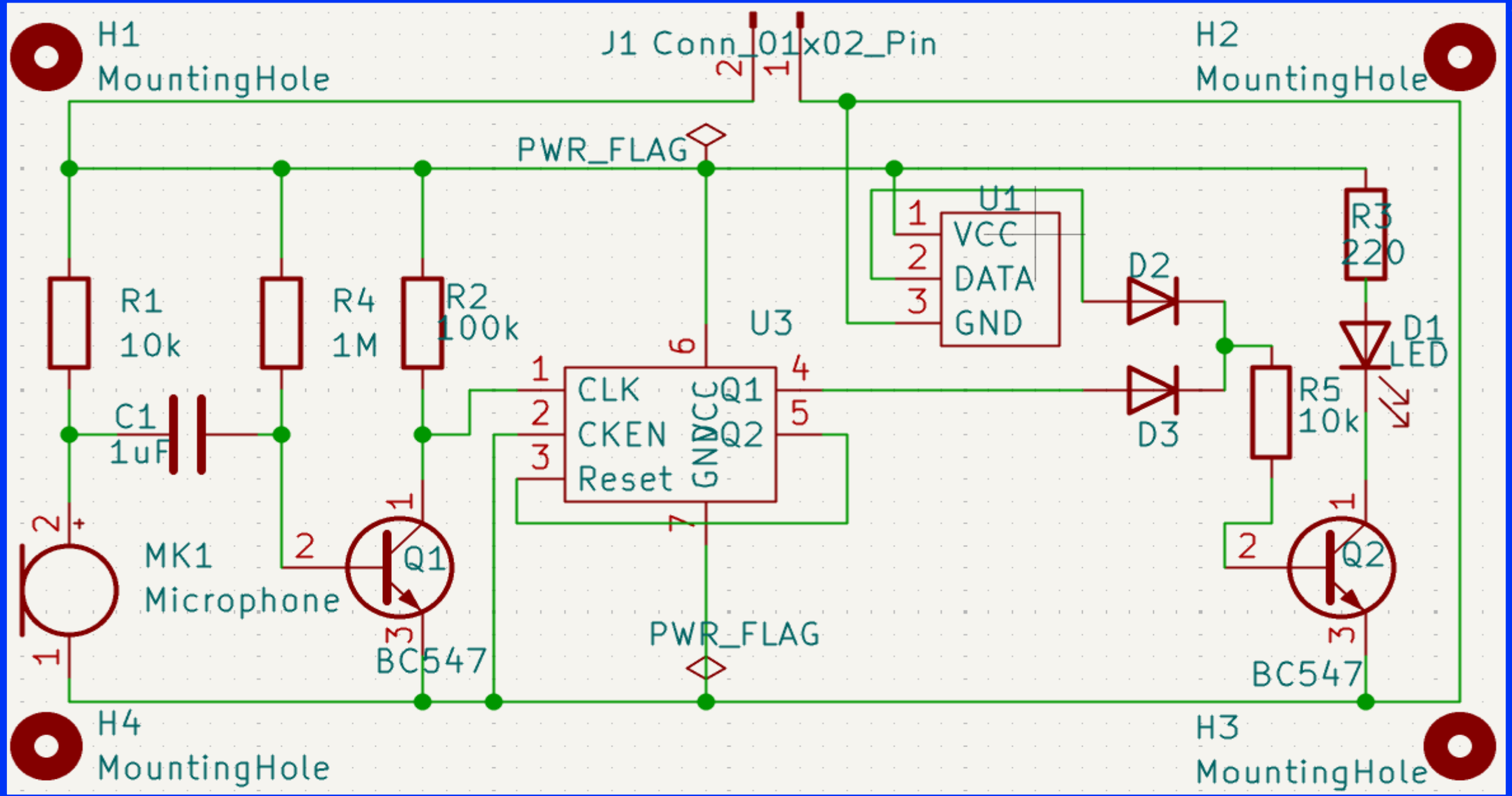
The complete circuit is developed as a PCB rather than being limited to a breadboard implementation. The PCB is designed as a minimum double-layer board, providing greater flexibility in routing the electrical connections.

OBJECTIVES

- The objectives of this project are:
- To design a clap-controlled electronic circuit.
- To detect clap sounds using a microphone.
- To amplify and condition the microphone signal.
- To use a 4017 decade counter for digital control.
- To use BC547 transistors as switching stages.
- To control an LED electronically.
- To incorporate a proximity sensing feature.
- To provide an LED indication when a person approaches the sensor.
- To design the complete circuit using KiCad.
- To assign appropriate PCB footprints to the components.
- To develop a minimum double-layer PCB.
- To perform PCB placement and routing.
- To perform electrical and design-rule checks.
- To generate a Bill of Materials.
- To generate Gerber manufacturing files.
- To gain practical experience in PCB design and manufacturing preparation.

CIRCUITARY MODEL (OR) SCHEMATIC DESIGN

**SCHEMATIC DESIGN
(OR)**



SYSTEM OVERVIEW

This project has two major functions.

- ✓ **Main function**

- Clap detection → Signal conditioning → 4017 U1 → Switching → LED

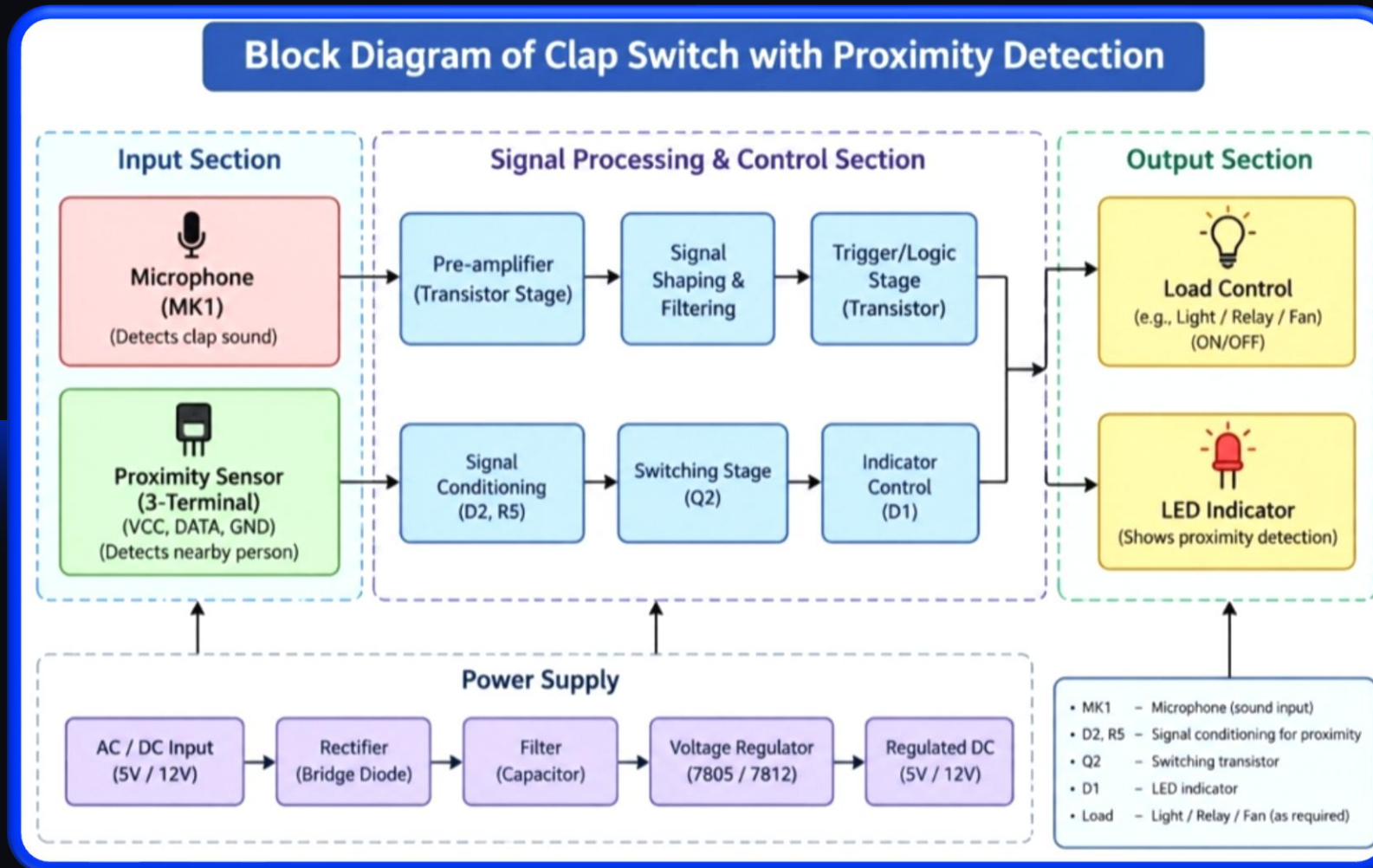
- ✓ **Additional function**

- Person approaches → Proximity sensor U3 → DATA signal → Diode/switching stage → LED

The final system therefore provides two ways of generating the LED indication:

- Clap-controlled operation
- Proximity-based operation

Block diagram



COMPONENTS USED

S.NO	Ref.	Component	Value/Part	Function
1	MK1	Microphone	Microphone	Clap Detection
2	Q1	NPN Transistor	BC547	Signal Processing
3	U3	Decade Counter	4017	Digital Control
4	Q2	NPN Transistor	BC547	Led Switching
5	D1	LED	LED	Output Indication
6	D2	Diode	Actual selected Diode	Proximity Signal Path
7	D3	Diode	Actual selected Diode	4017 Signal Path
8	R1	Resistor	10k Ω	Microphone Circuit

COMPONENTS USED

S.NO	Ref.	Component	Value/Part	Function
9	R2	Resistor	100k Ω	Bias/Signal Network
10	R3	Resistor	220 Ω	LED Current Limiting
11	R4	Resistor	1M Ω	Input Bias Network
12	R5	Resistor	10k Ω	Q2 Base Drive
13	C1	Capacitor	1 μ F	signal Filtering
14	J1	Connector	2-pin	Power Input
15	Sensor	Proximity Module	VCC/DATA/GND	Person Detection
16	H1-H4	Mounting Holes	-	PCB Mounting

THEORETICAL BACKGROUND

✓ Sound Detection

A clap produces a sudden acoustic pressure wave.

The microphone converts this acoustic energy into a small electrical signal.

The electrical signal is then processed by the transistor stage.

✓ Microphone

The microphone MK1 acts as the sound sensor.

When a clap occurs, the microphone generates a corresponding electrical signal.

Because the signal produced by a microphone is small, additional circuitry is required before it can be used to drive the digital counter.

✓ BC547 Transistor

The BC547 is an NPN bipolar junction transistor.

It has three terminals:

- Collector
- Base
- Emitter

In this project:

Q1

Q1 is used in the microphone signal-processing section.

The microphone signal is applied to the base region through the associated coupling/bias network.

The transistor produces a switching/amplified signal suitable for the following digital stage.

Q2

Q2 is used as the output switching transistor.

Its base receives the control signal through $R5 = 10\text{ k}\Omega$.

When sufficient base drive is applied, Q2 conducts and controls the current through the LED.

✓ 4017 Decade Counter

The 4017 is a decade counter with decoded outputs.

It changes its active output when a clock pulse is applied.

In the project:

- Clap → Q1 → Clock input of 4017 → Output changes

The counter therefore converts the detected clap pulses into sequential digital output states.

The 4017 is supplied from the circuit's positive supply and ground.

In your schematic, the relevant signals are:

- CLK
- CKEN
- RESET
- Q1
- Q2
- VCC
- GND

✓ LED

- D1 is the visual output indicator.
- The LED is connected with $R3 = 220 \Omega$ as its current-limiting resistor.
- The purpose of $R3$ is to limit the current through the LED and protect the LED from excessive current.
- The basic LED resistor equation is:

$$R = (VCC - V_F) / I_F$$

where:

- VCC = supply voltage=5V
- V_F = LED forward voltage=around 2V
- I_F = LED current=15mA
- $R3$ = required series resistor(220Ω)

✓ Diodes

- D2
- D3

D2 and D3 are used in the control/output section.

They allow the outputs of different sections of the circuit to be combined while helping prevent unwanted interaction between those signals.

In circuit:

- D2 receives the DATA signal from the proximity sensor.
- D3 receives the relevant 4017 control output.
- Their outputs are combined and applied toward the Q2 switching stage through R5.
- This allows both the proximity detection section and the clap-control section to influence the LED output.

✓ Resistors

schematic contains:

- $R1 = 10\text{ k}\Omega$
- Used in the microphone/input biasing network.
- $R2 = 100\text{ k}\Omega$
- Used in the signal-conditioning/bias network associated with Q1.
- $R3 = 220\text{ }\Omega$
- Used as the LED current-limiting resistor.
- $R4 = 1\text{ M}\Omega$
- Used in the microphone/transistor input bias network.
- $R5 = 10\text{ k}\Omega$
- Used to limit/control the base drive to Q2.

✓ Capacitor C1

- $C1 = 1\text{ }\mu\text{F}$
- C1 is used for signal coupling/filtering in the microphone section.
- It helps transfer the changing clap signal while providing the required DC isolation between the associated circuit nodes.

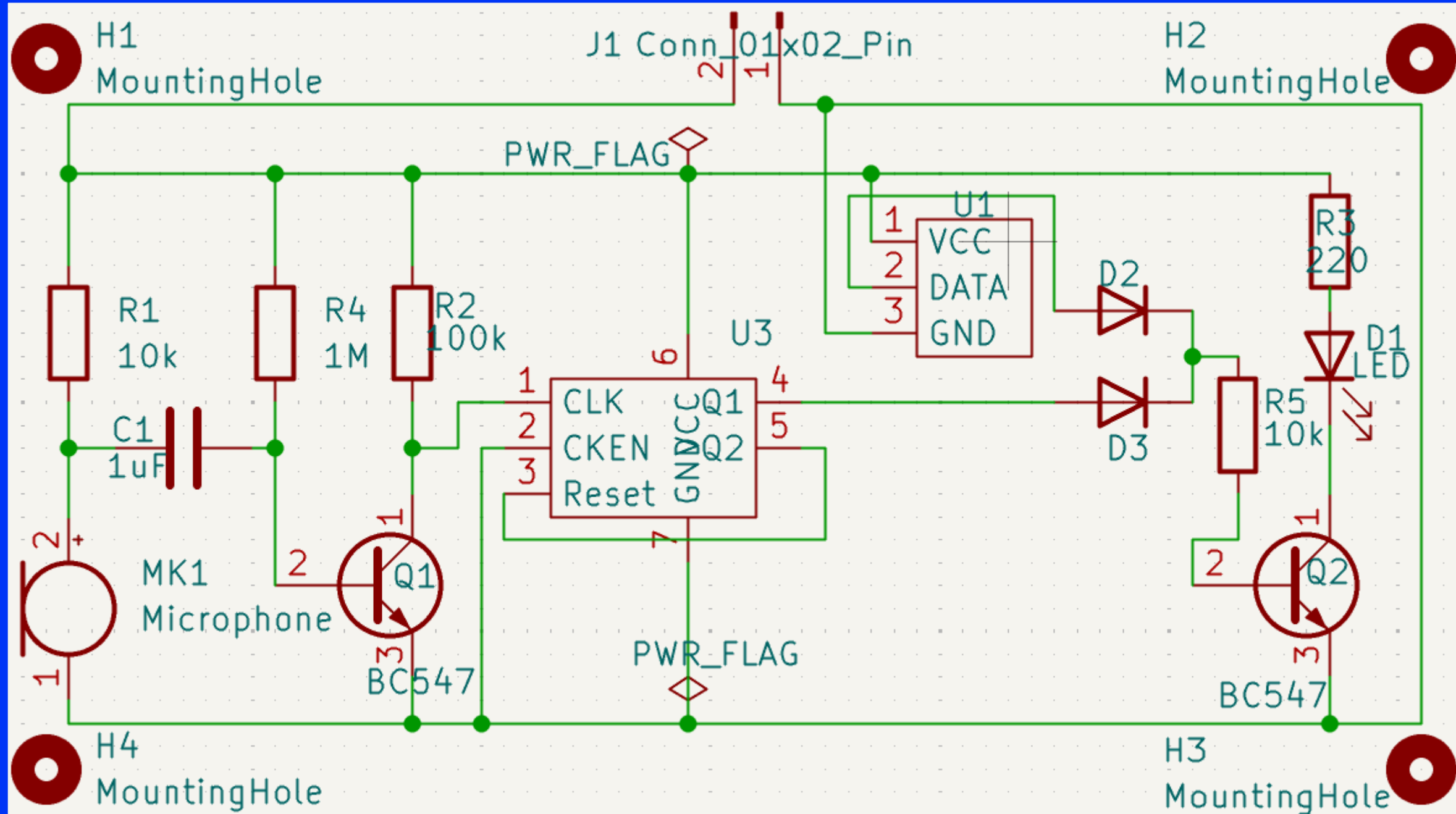
✓POWER SUPPLY

The circuit is designed around a low-voltage DC supply.

J1 is the two-pin power connector.

The power connections are:

- J1 → VCC/+5 V
- J1 → GND
- The VCC line supplies the active circuit components, while the GND line provides the common return path.
- PWR_FLAG symbols are included in the schematic to indicate valid power-driving nets to KiCad's electrical-rule checking system.



CIRCUIT WORKING

The working of the project can be divided into two sections.

➤ Clap-controlled operation :

- When a person claps near the microphone:
 - ✓ The sound reaches MK1.
 - ✓ MK1 converts the sound into an electrical signal.
 - ✓ The signal is coupled through C1.
 - ✓ The signal is applied to the Q1 transistor stage.
 - ✓ Q1 processes the signal and produces a switching pulse.
 - ✓ The pulse is applied to the CLK input of U3.
 - ✓ The 4017 changes its active output.
 - ✓ The relevant output passes through D3.
 - ✓ The signal reaches the Q2 base through R5.
 - ✓ Q2 switches.
 - ✓ Current flows through D1.
 - ✓ The LED turns ON/OFF according to the designed counter state.

ADDITIONAL FEATURE
IS
“Proximity Detection Operation”

“Proximity Detection Operation”

IS

✓ Proximity Detection Operation

The **ADDITIONAL FEATURE** uses a three-terminal proximity sensor/module.

The sensor has:

- VCC
- DATA
- GND

The DATA output is connected to the control section through D2.

□ When a person approaches the sensing region:

Person → Sensor → DATA output → D2 → R5 → Q2 → D1

➤ Therefore, the LED provides a visual indication of the detected person.

Project-specific feature

The system is intended to provide the LED indication when a person approaches approximately 1 meter from the sensor.

However, the actual detection distance depends on:

- The sensor/module used
- Sensor adjustment
- Environmental conditions
- Target size
- Target reflectivity
- Sensor orientation

So, during physical testing, measure and record the actual detection distance.

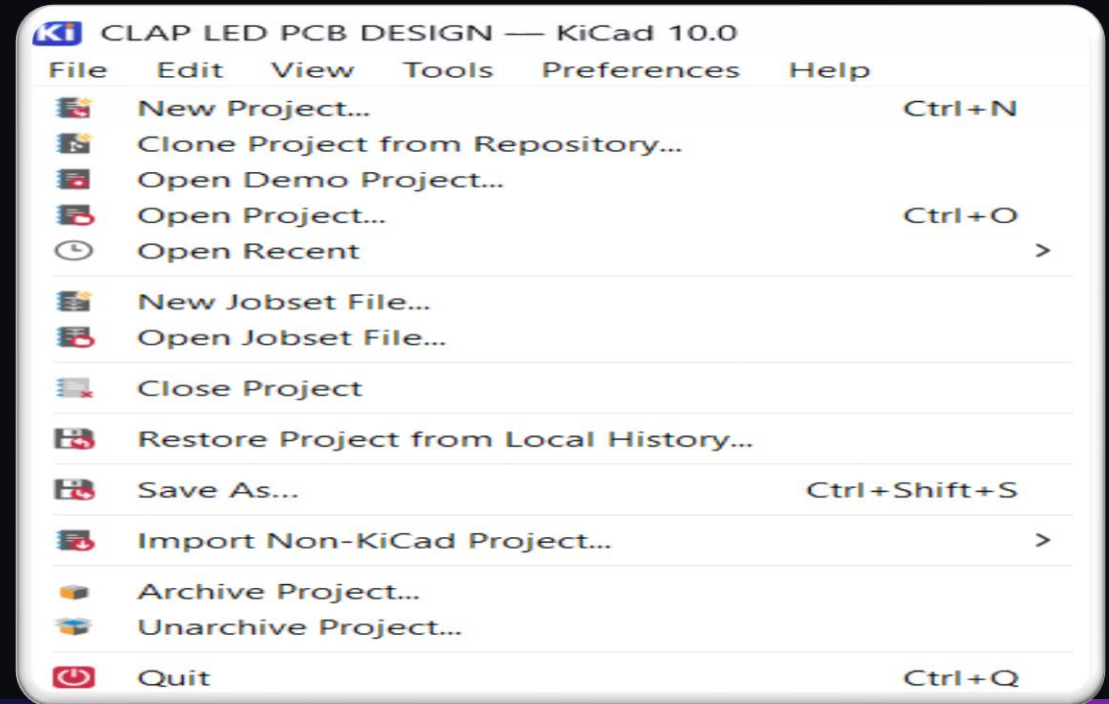
SCHEMATIC DESIGN
USING

KICAD

KICAD

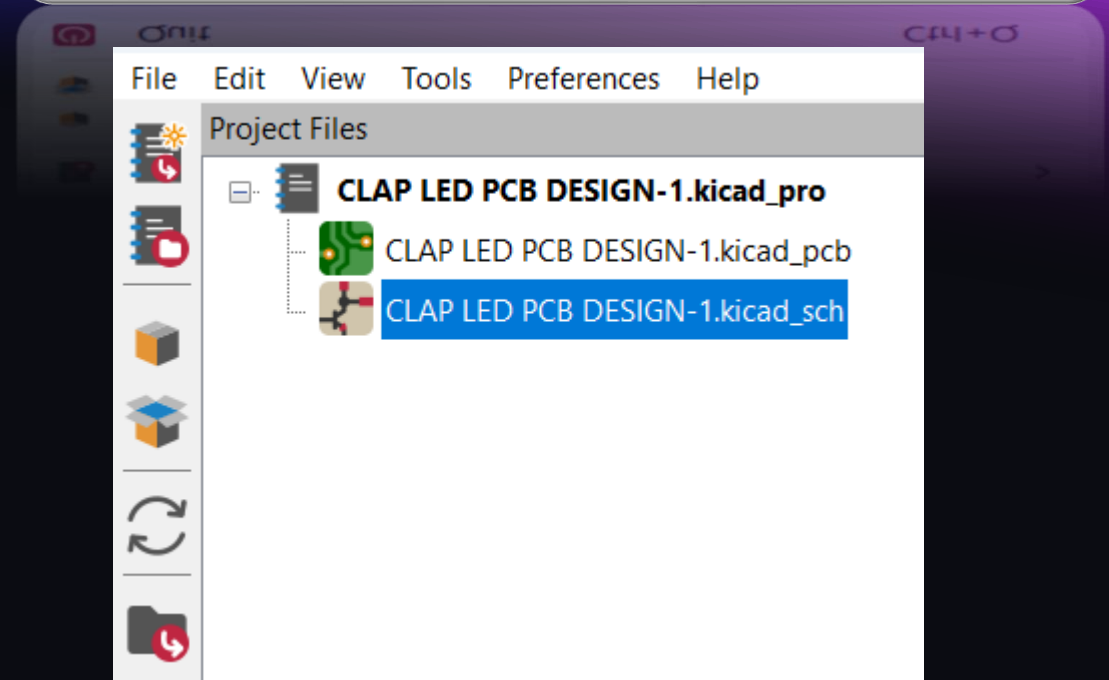
Step 1: Create project

Open KiCad and create a new project.



Step 2:

- Name the File
- Select (File Name).kicad_sch

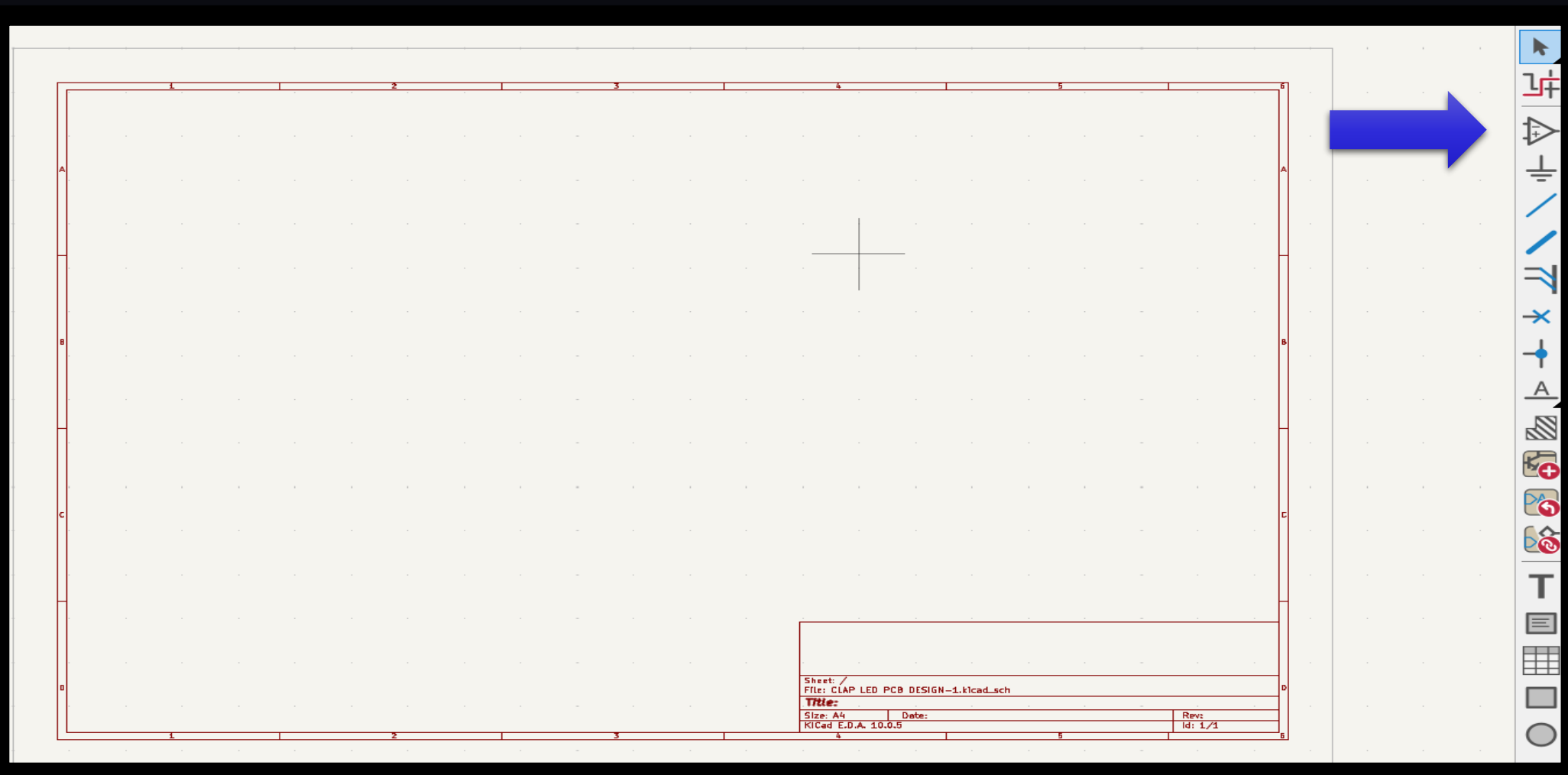


Step 3:

Select



right corner



Step 4: Component Placement

The following components were placed:

- Microphone
- BC547 transistors
- LEDs
- Diodes
- Resistors
- Capacitor
- Power connector
- Power symbols
- Ground symbol

Note :(search all these components in the search button one by one in schematic Editor)

Step 5 :(Exception)

Since our moto is to make a PCB as very compact (small) so, we will make an:

- 1) IC(4017)
- 2) SENSOR(HC-SR04 Proximity Sensor) as our own

Step 6 :

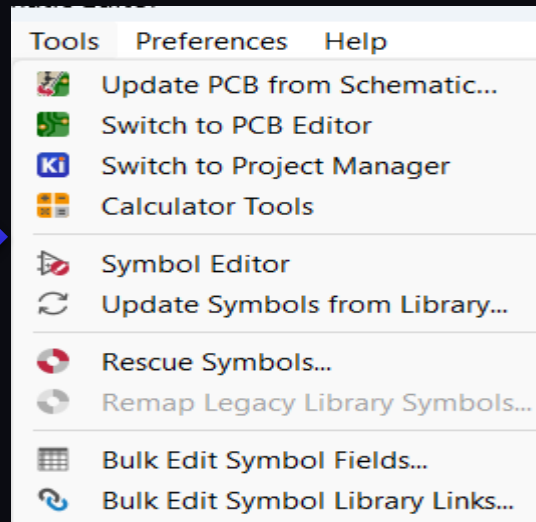
On Top select TOOLS button to create our own

- IC
- SENSOR

STEP 7:

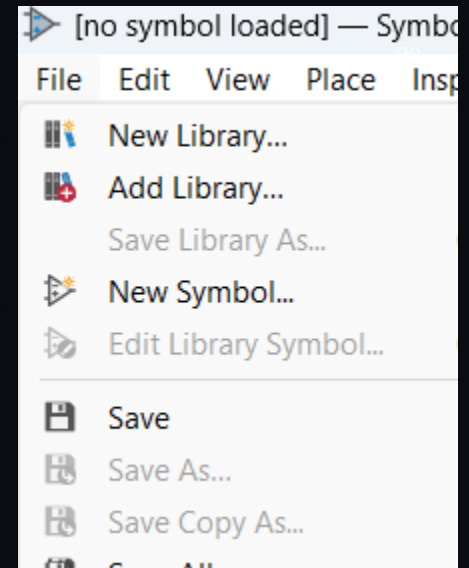
Choose :

- Symbol Editor



Step 8:

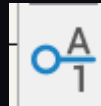
- Go to File
- New Library
- Save the file
- Go to New symbol
- Click ok



Step 9:

✓ On Right side :

- select



- Remember this symbol (will use later)



Step 10:

• NOTE :

❑ To Create an IC we need 5pins in the project...

➤ Those are:

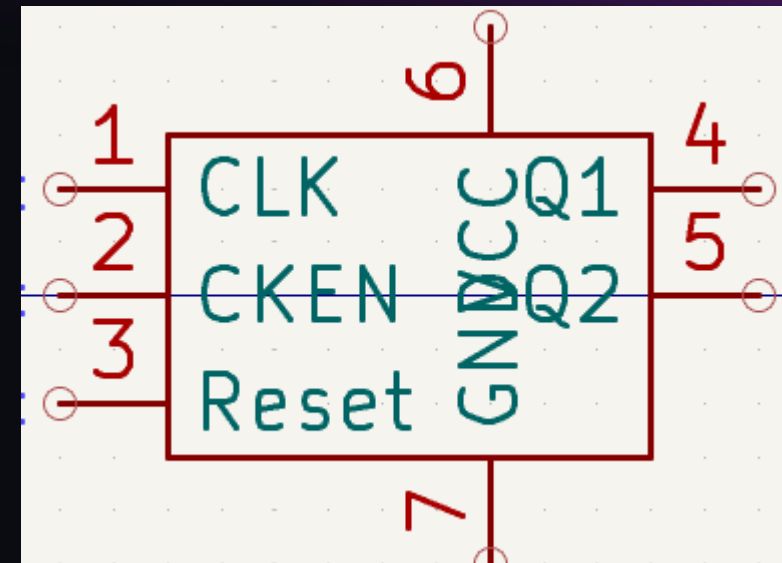
1. CLK (input)
2. CKEN (input)
3. RESET (input)
4. Q1 (output)
5. Q2 (output)
6. VCC (input power)
7. GND (input power)

Put the same measurements to all these 5
And place the result one by one in order ...

8. Place the Rectangular box which is in the right corner



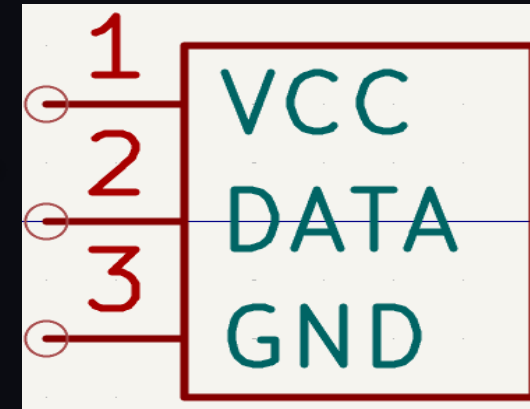
Pin Properties	
Pin name:	CLK
Pin number:	1
Electrical type:	→ Input
Graphic style:	└ Line
X position:	450 mils
Y position:	150 mils
Orientation:	○ Right
Pin length:	100 mils
Name text size:	50 mils
Number text size:	50 mils



Step 11:

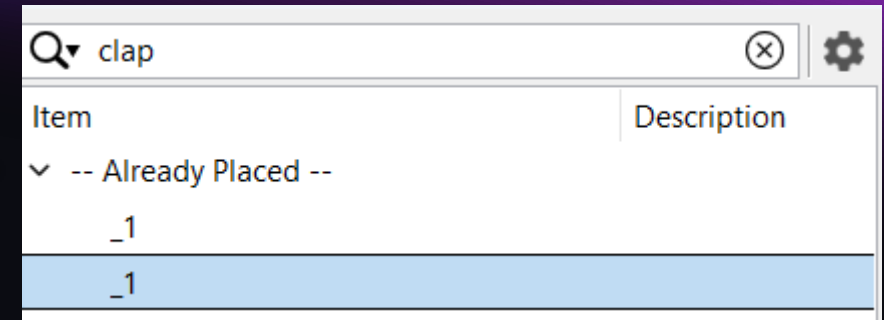
Similarly do for HC-SR04 SENSOR in the same method but needs only 3pin i.e.,

- VCC
- DATA
- GND



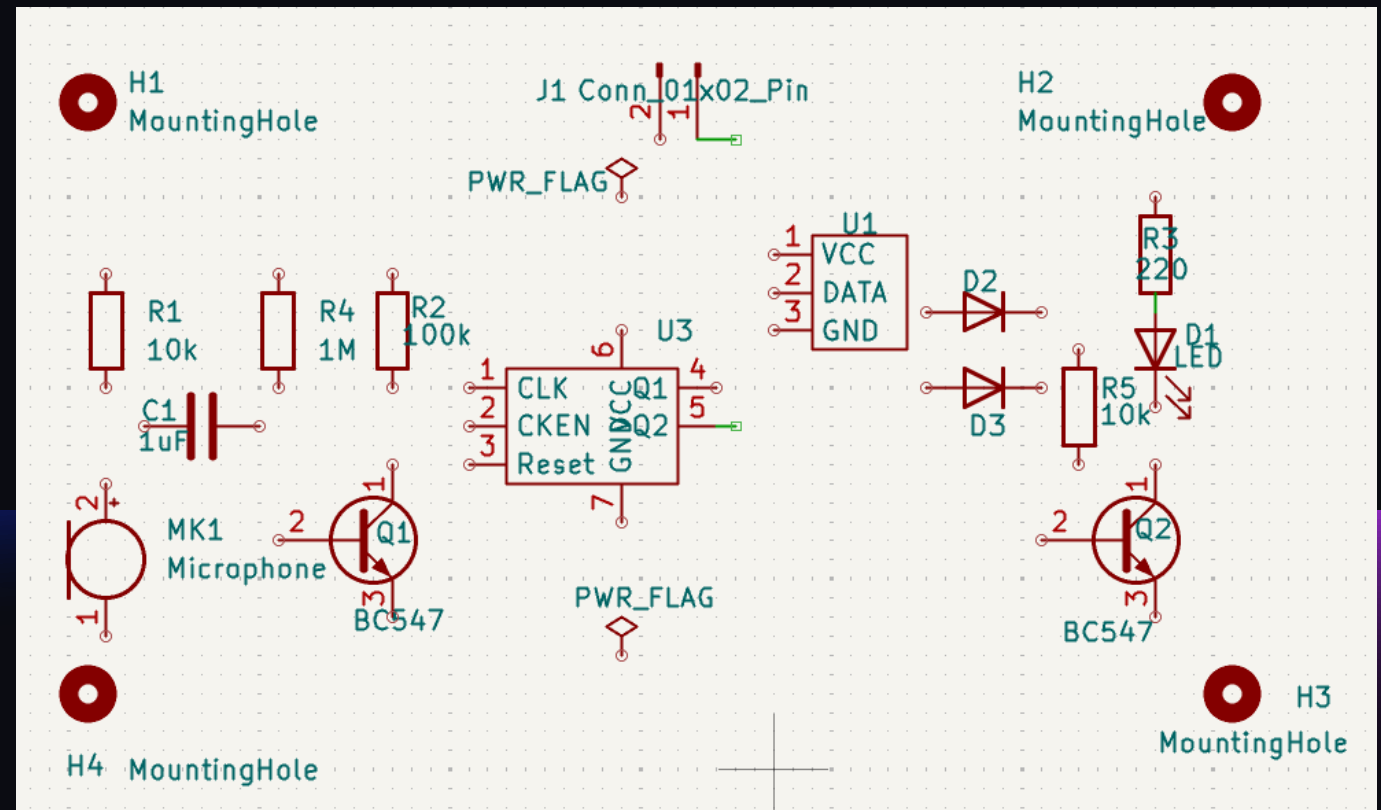
Step 12:

- Now come back to Schematic and press A
- Then Type your file Name that you saved ...
- Press ok



Step 13:

- Now place all the Components in SCHEMATIC EDITOR and Arrange them exact places



Circuit Connections

The components were connected according to Below:

- The major connections include:
 - ✓ Microphone section
 - VCC → R1 → Microphone
 - Microphone → C1 → Q1 input
 - Q1 section
 - Q1 is connected to the microphone signal-processing network.
 - ✓ Counter section
 - Q1 output → 4017 CLK
 - Proximity section
 - Sensor DATA → D2
 - Counter output
 - 4017 output → D3
 - ✓ Combined control
 - D2 + D3 → R5 → Q2 base
 - LED section
 - VCC → R3 → D1 → Q2
 - Q2 → GND

POWER AND GROUND

✓ Power Connections

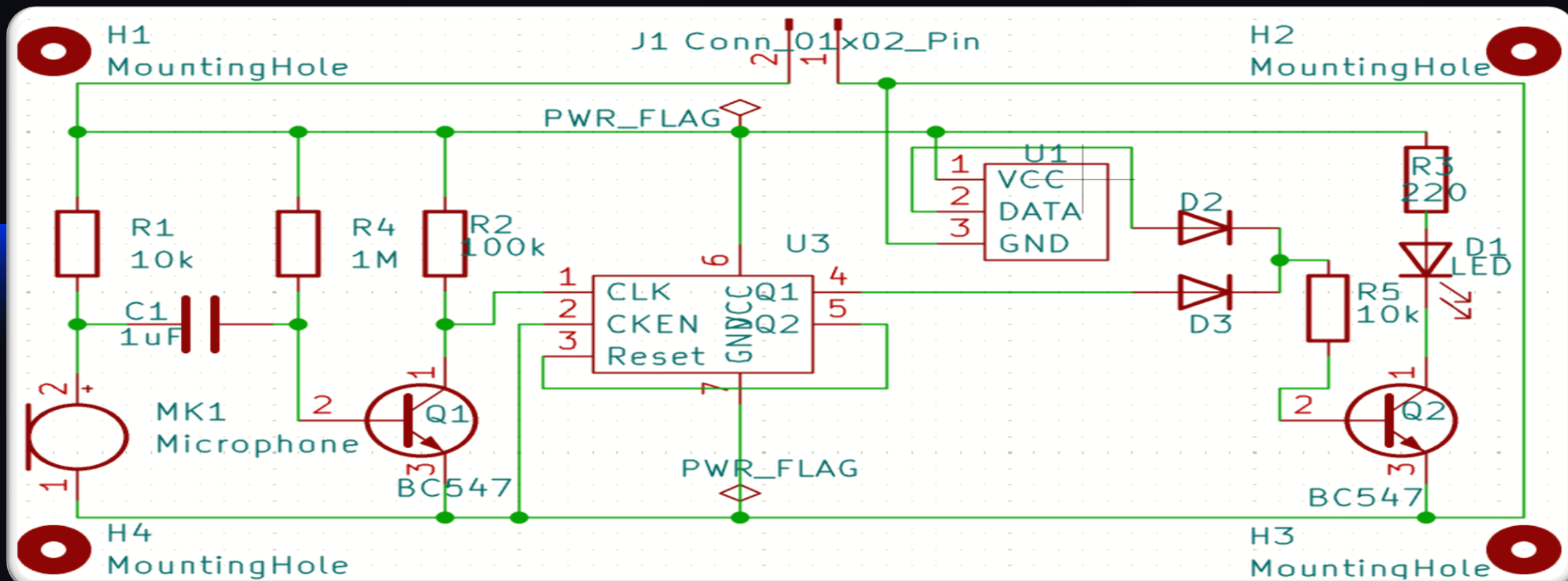
The required power connections were added.

□ The schematic contains:

- VCC
- GND
- PWR_FLAG
- The power network supplies the 4017, sensor and other required circuit sections.

Step 14:

- Now connect them Like this
- Your Result should be as



FOOTPRINT ASSIGNMENT

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- The schematic symbols represent the electrical components, but PCB design requires their physical package dimensions.
- Therefore, suitable footprints were assigned.

Step 14:

- Go to each symbol and Press F on it
- Search the Footprint
- Click ok



Edit Footprint Field

Footprint:

☐ Visible ☐ Show field name ☒ Allow automatic placement

Font: **B** /        

Text size: mils Color: 

Position X: mils

Position Y: mils

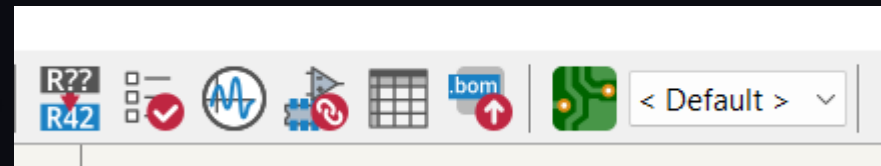
SCHEMATIC VERIFICATION

Step 14:

On the Top click on this



Symbol to verify ERC



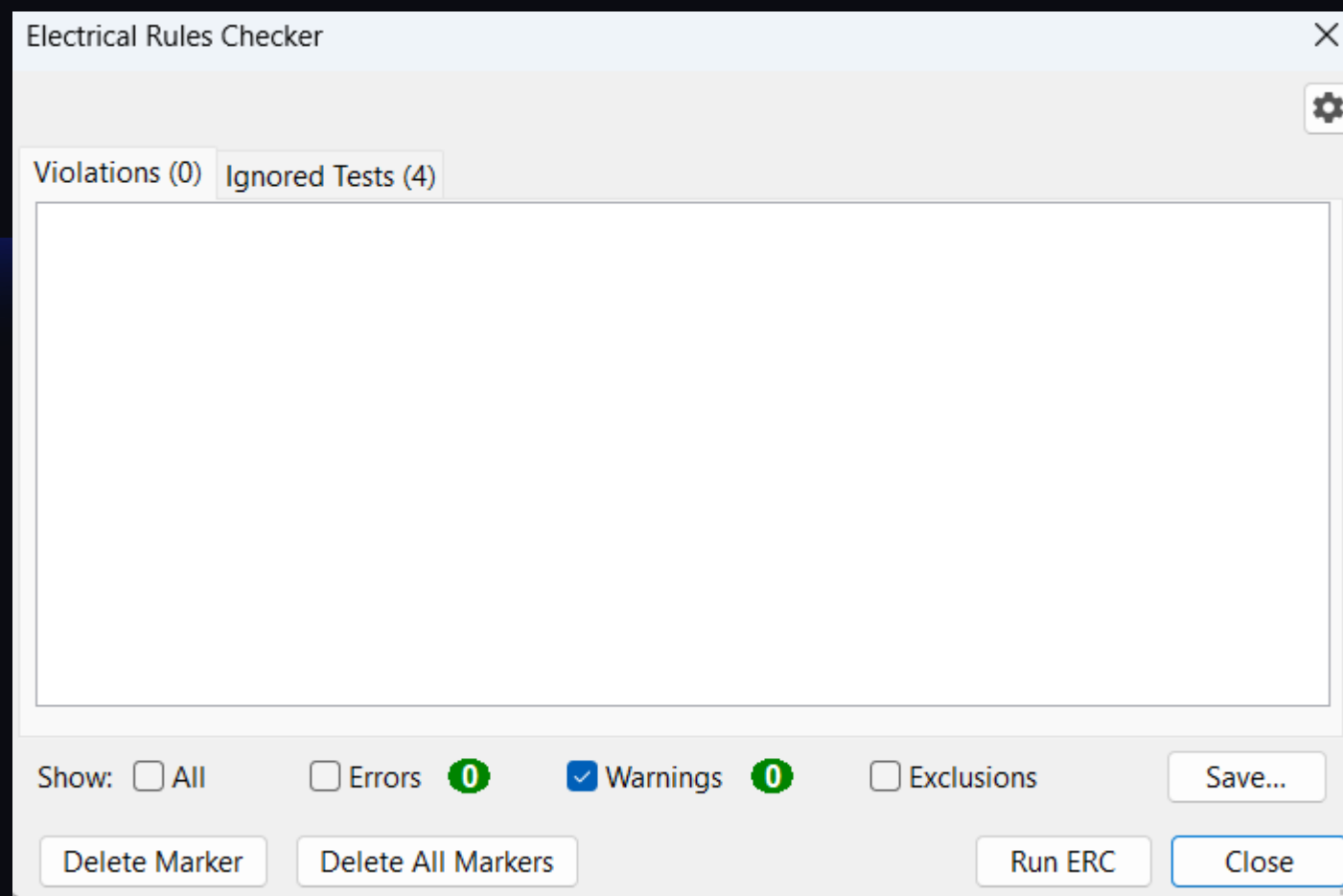
Step 15:

ELECTRICAL RULE CHECK

After completing the schematic, the Electrical Rules Checker was performed.

❑ ERC was used to identify:

- Unconnected pins
- Incorrect electrical connections
- Power-related warnings
- Unconnected nets
- Other schematic design issues



BOM(Bill Of Materials)

- BOM is a complete list of all components required to assemble a PCB.
- It contains reference designator, component name, value, footprint/package, and quantity.
- It may also include the manufacturer and part number of each component.
- It is used for purchasing and organizing all components needed for PCB assembly.
- It helps to avoid missing/wrong components and estimate the total cost of the PCB.

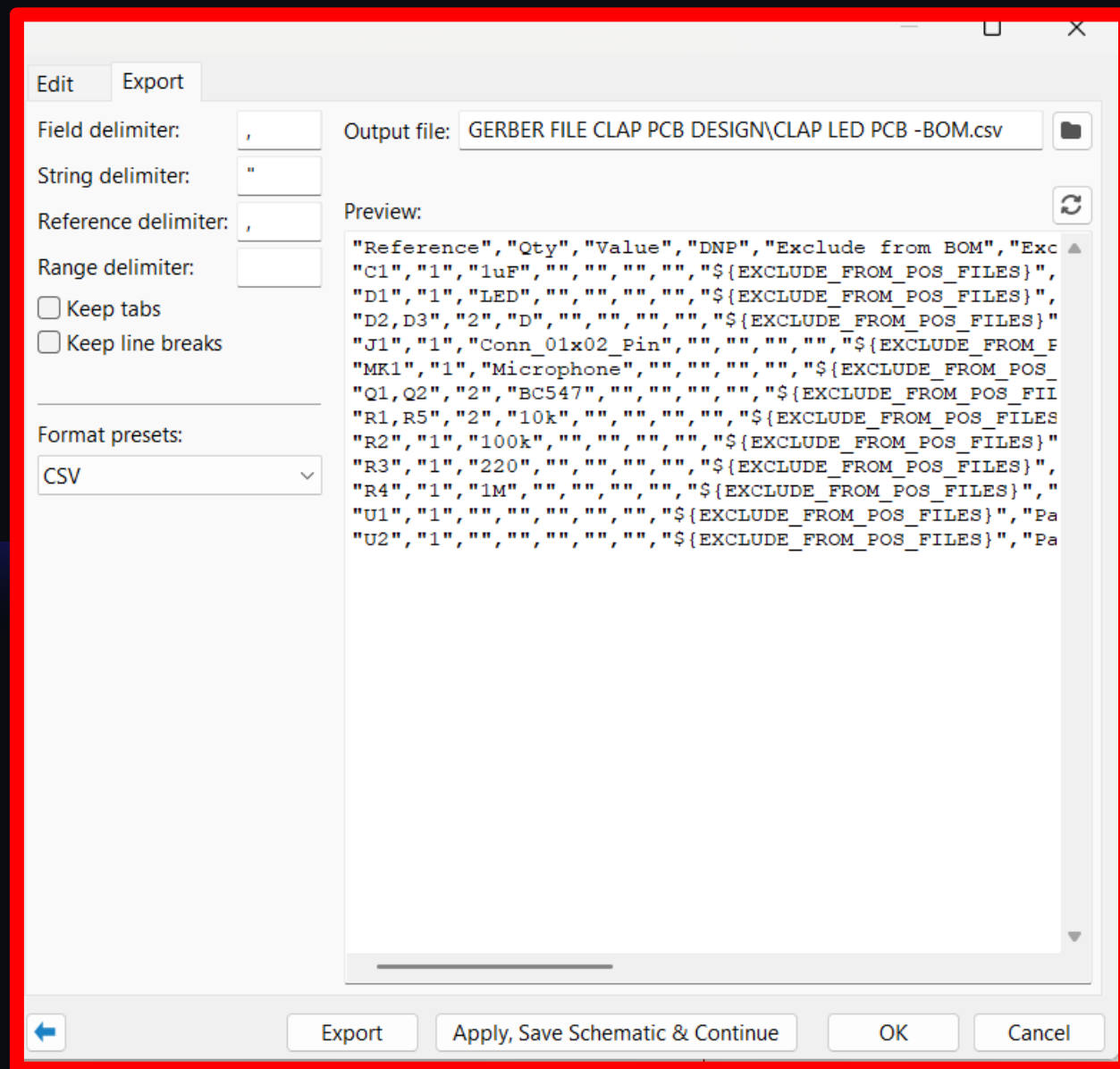
Step 16 :

- Select At Top right a Symbol called as BOM



Step 17 :

- Select the file
- Press Apply , Save, Schematic and continue button
- & Export

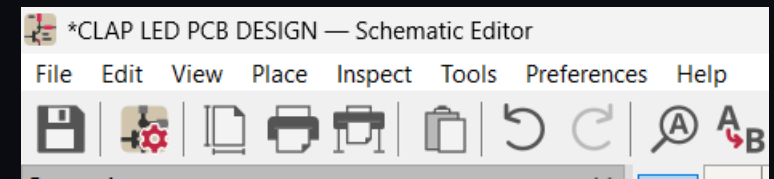


PCB DESIGNING

PCB DESIGNING

Step 19:

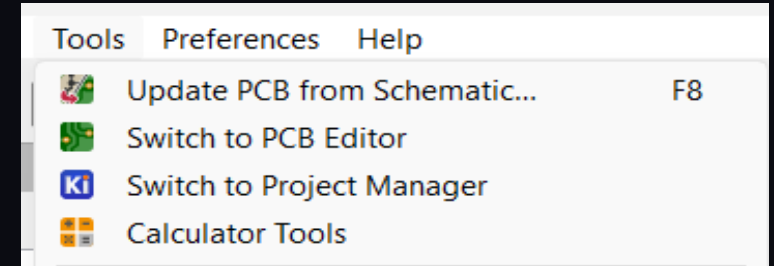
Select the option Tools in schematic only at Top



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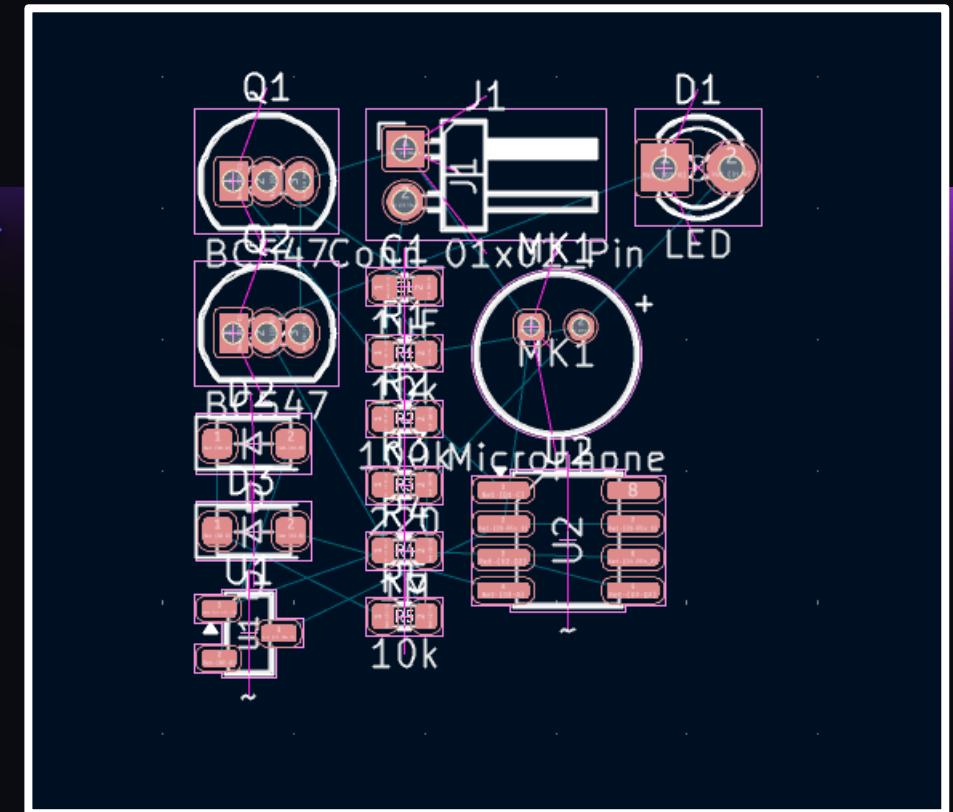
Step 20:

Select Update PCB from SCHEMATIC



Step 21:

After Updating PCB from schematic ,will be getting like this



Step 22 :

Since to reduce the size of PCB,
Will be using both back and front.

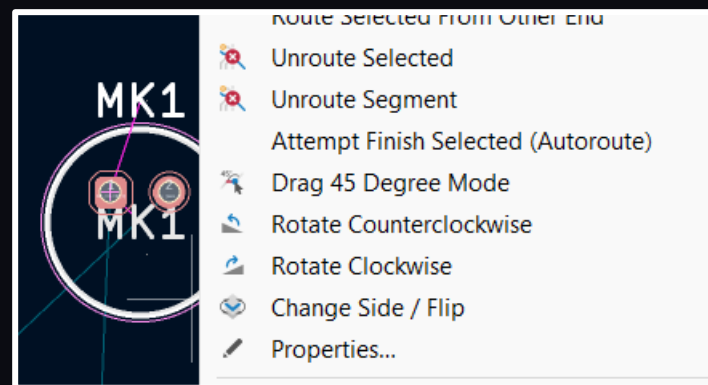
So for that to shift the devices to back ,Follow these Steps:



EXAMPLE:

To Flip the device back side of PCB,DO Right click on whatever then object you need to Flip.

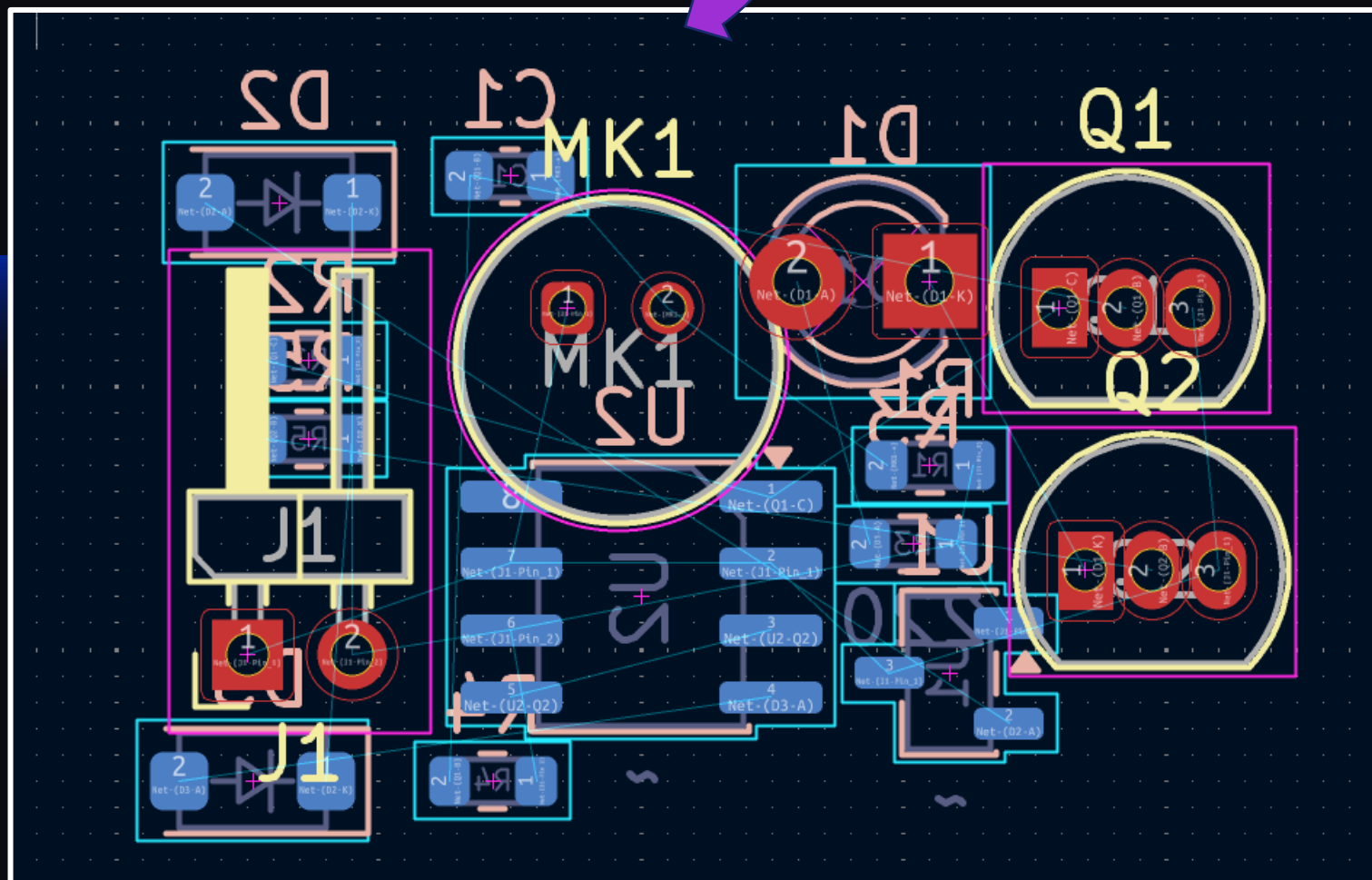
Then choose the option Change Side/Flip.
Now the object is Flipped to backside.



Step 23:

Arrange the devices where ever you need , but our motto is to make PCB as much as small.

So After Rearranging you see a Result like this:



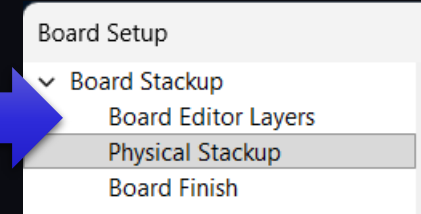
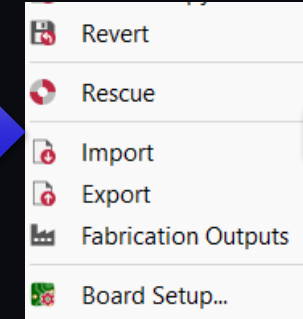
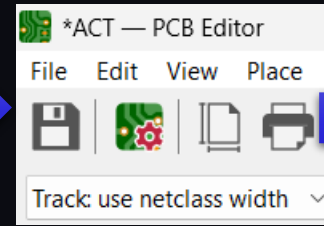
MULTILAYER - PCB

MULTILAYER - PCB

Step 24:

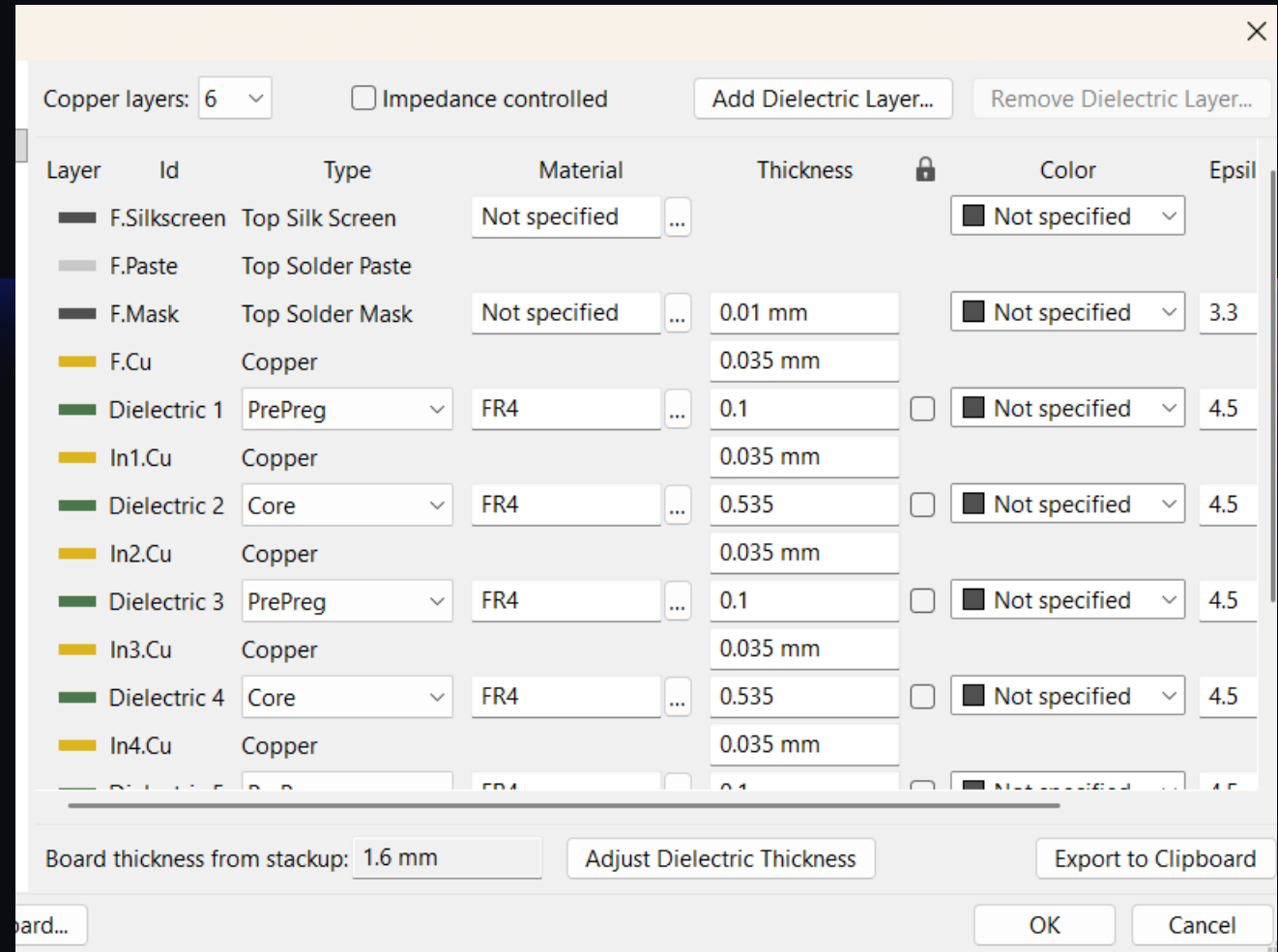
On Top Left :

1. Open File
2. Select Board setup
3. Physical stack up



Step 25:

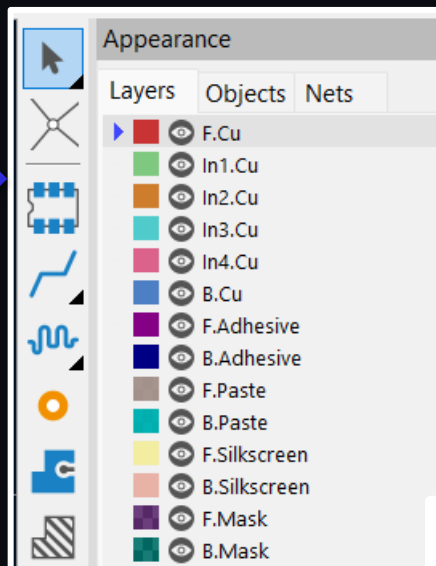
Choose How many Layers of PCB do we need and select ok.(I choose 6)



Step 26:

Now we get all Layers.

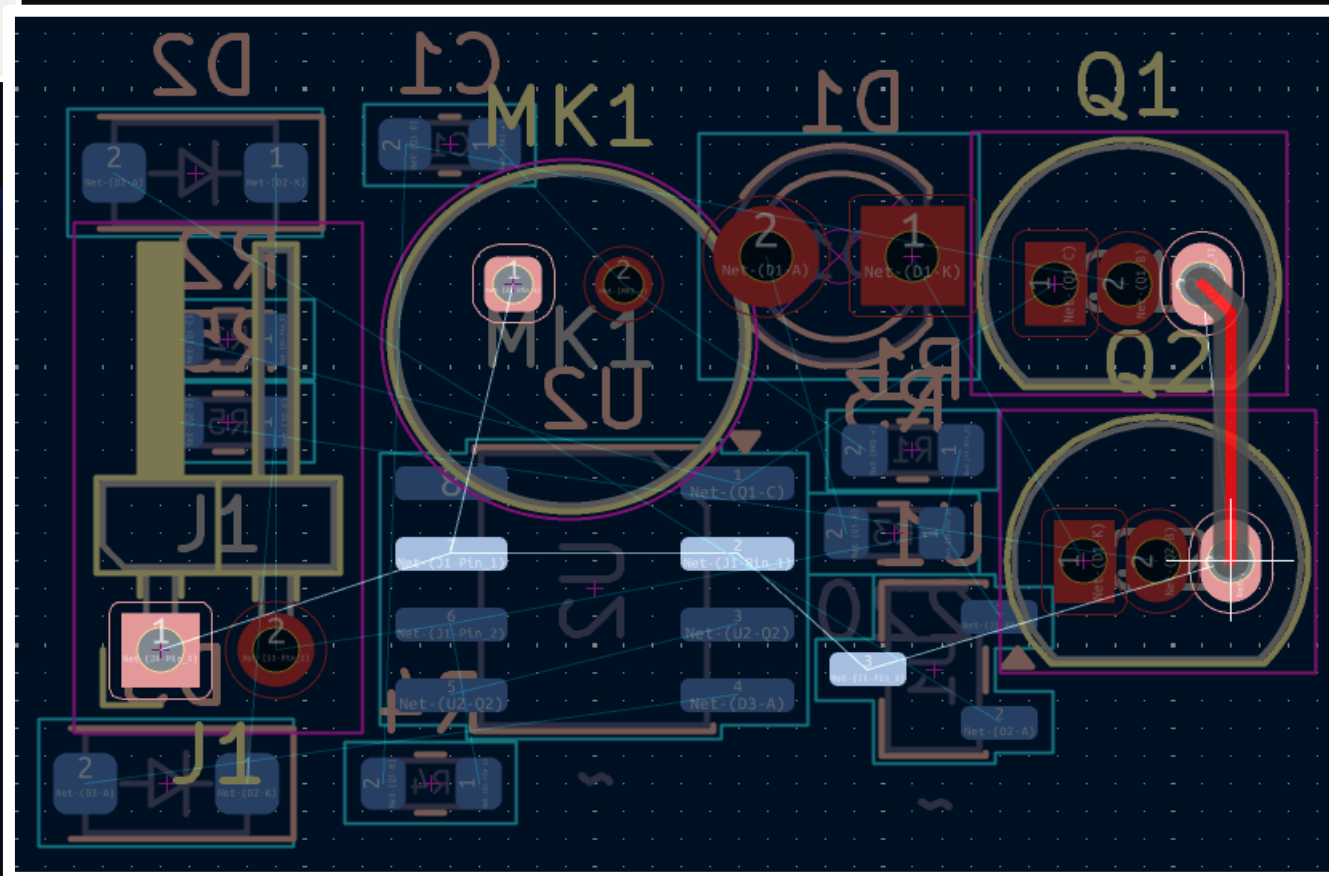
1. F.Cu
2. In1.Cu
3. In2.Cu
4. In3.Cu
5. In4.Cu
6. B.Cu



Step 27:



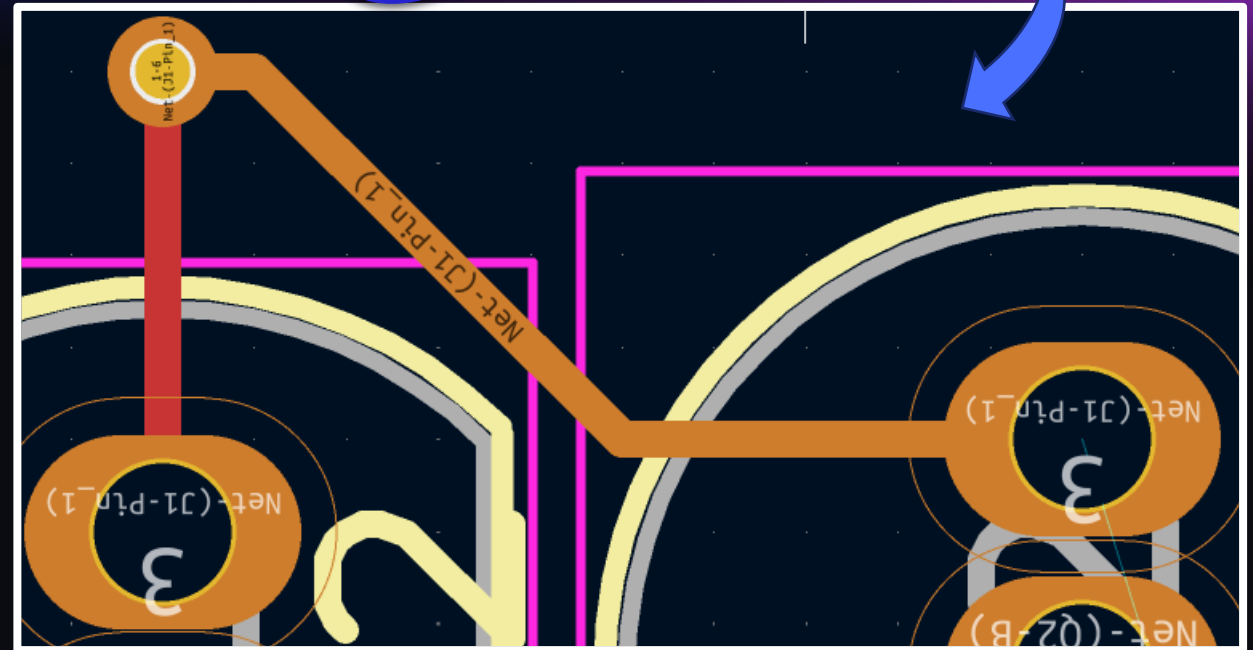
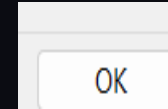
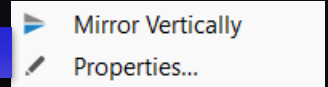
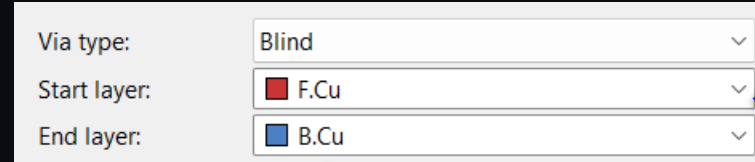
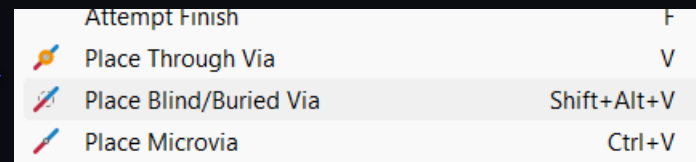
This symbol is to be used while Joining the devices



Step 28:

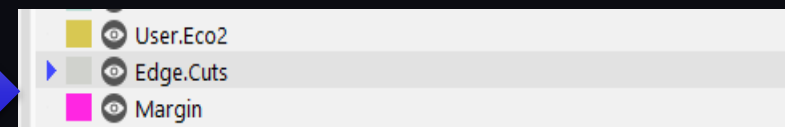
To get MULTILAYER Connection will be using BURRIED VIAS is as FOLLOWS:

- Right click at middle of connection while joining
- Select Place Blind/Buried VIA
- Choose arrow to change the LAYER
- Again Right click on this
- Select Properties
- Choose the LAYERS that you need
- Confirm ok
- Connect the other device



Step 29:

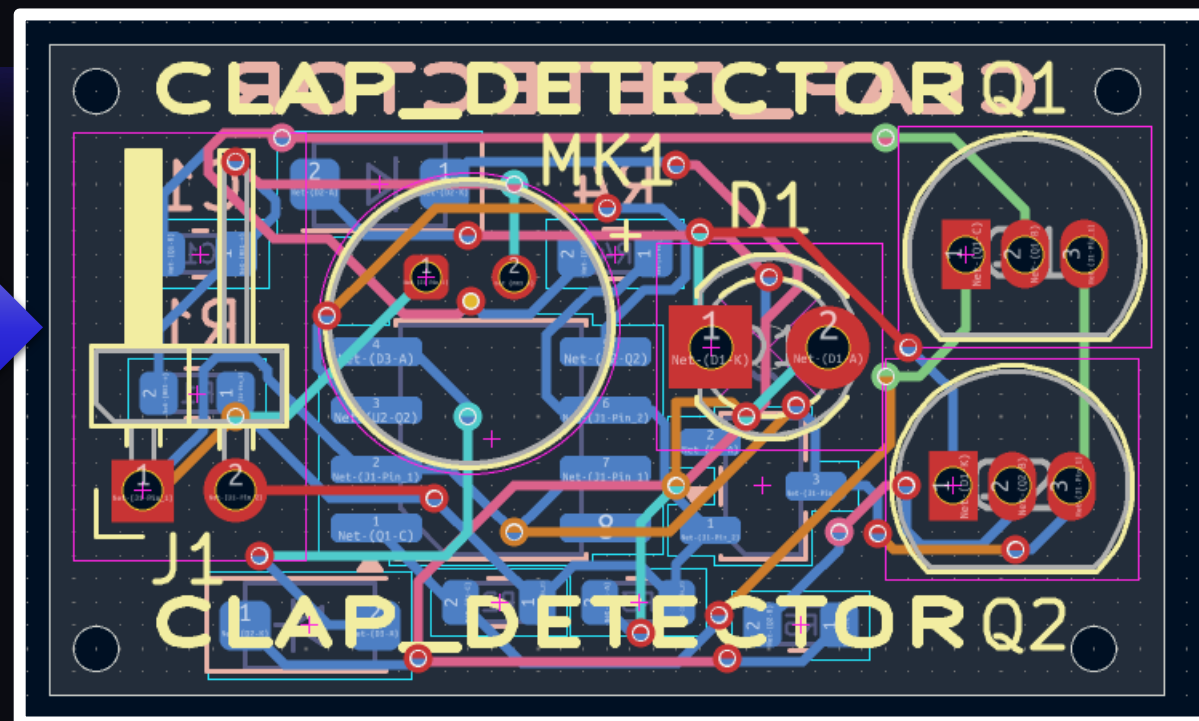
- Select an Egdecut at Right Corner and draw as (Border)
- Keep Edge cut holes at corners
- If possible, write Text format i.e. , CLAP_DETECTOR



Step 30 :

Connect all the Devices as guided as above steps.

Output may Look like :



VERIFICATION (DRC)

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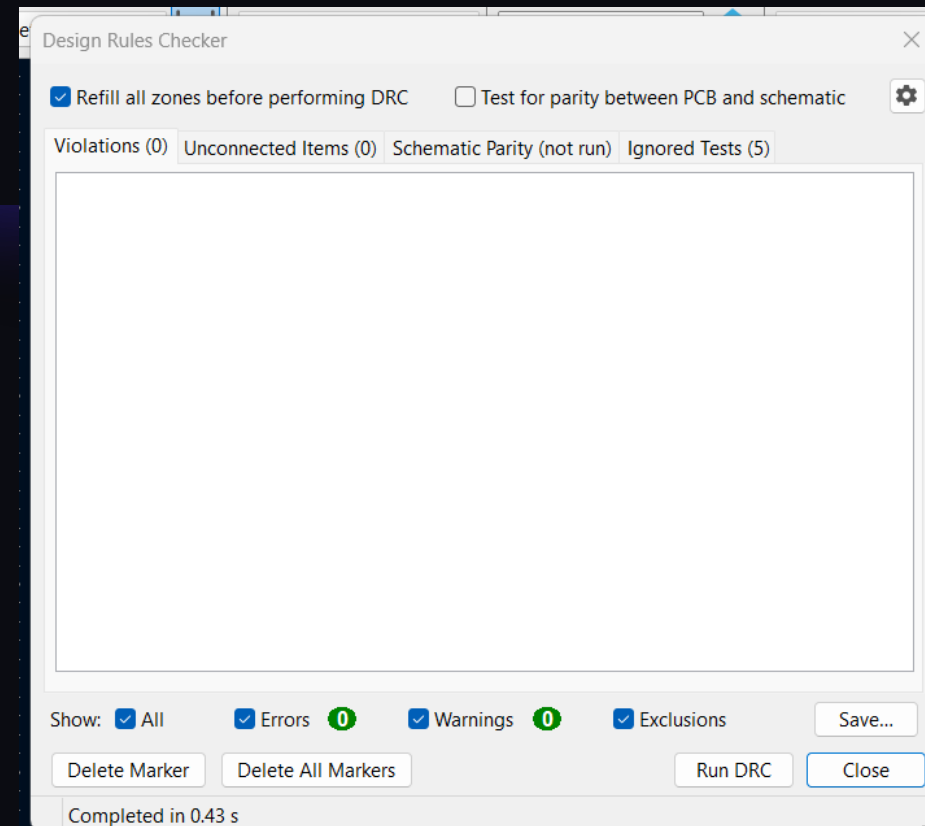
Step 31:



Select this Symbol to check are there any warnings OR errors are present By running DRC

Therefore,

- ✓ NO ERRORS
- ✓ NO WARNINGS



HOW TO MAKE GERBER FILE ?

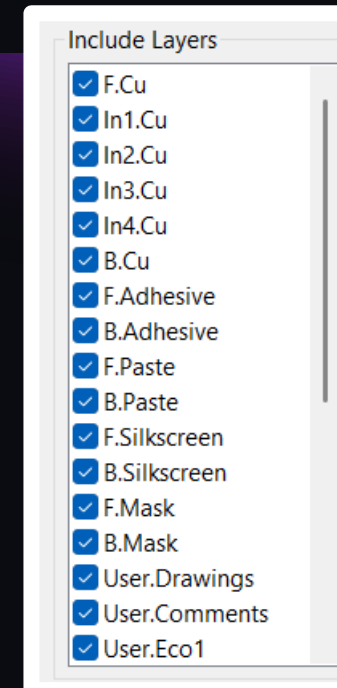
Step 32 :



- ❖ Select this symbol on Top bar to save all files as GERBER.

Step 33:

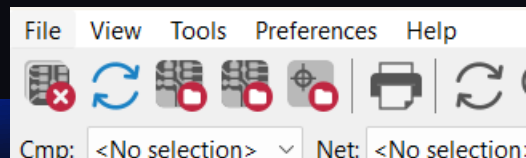
- ❖ Select all these Layers
- ❖ Save the File where ever you want to save
- ❖ Plot and generate all Drill Files into same saved Filed
- ❖ And Zip that Complete Gerber file



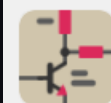
HOW TO OPEN GERBER FILE ?

Step34:

- Select Gerber Viewer
- On top Select File



& Open your Zipped GERBER file that you saved in Files.



Schematic Editor

Edit the project schematic



Symbol Editor

Edit global and/or project schematic symbol libraries



PCB Editor

Edit the project PCB design



Footprint Editor

Edit global and/or project PCB footprint libraries



Gerber Viewer

Preview Gerber files



Image Converter

Convert bitmap images to schematic symbols or PCB footprints



Calculator Tools

Show tools for calculating resistance, current capacity, etc.



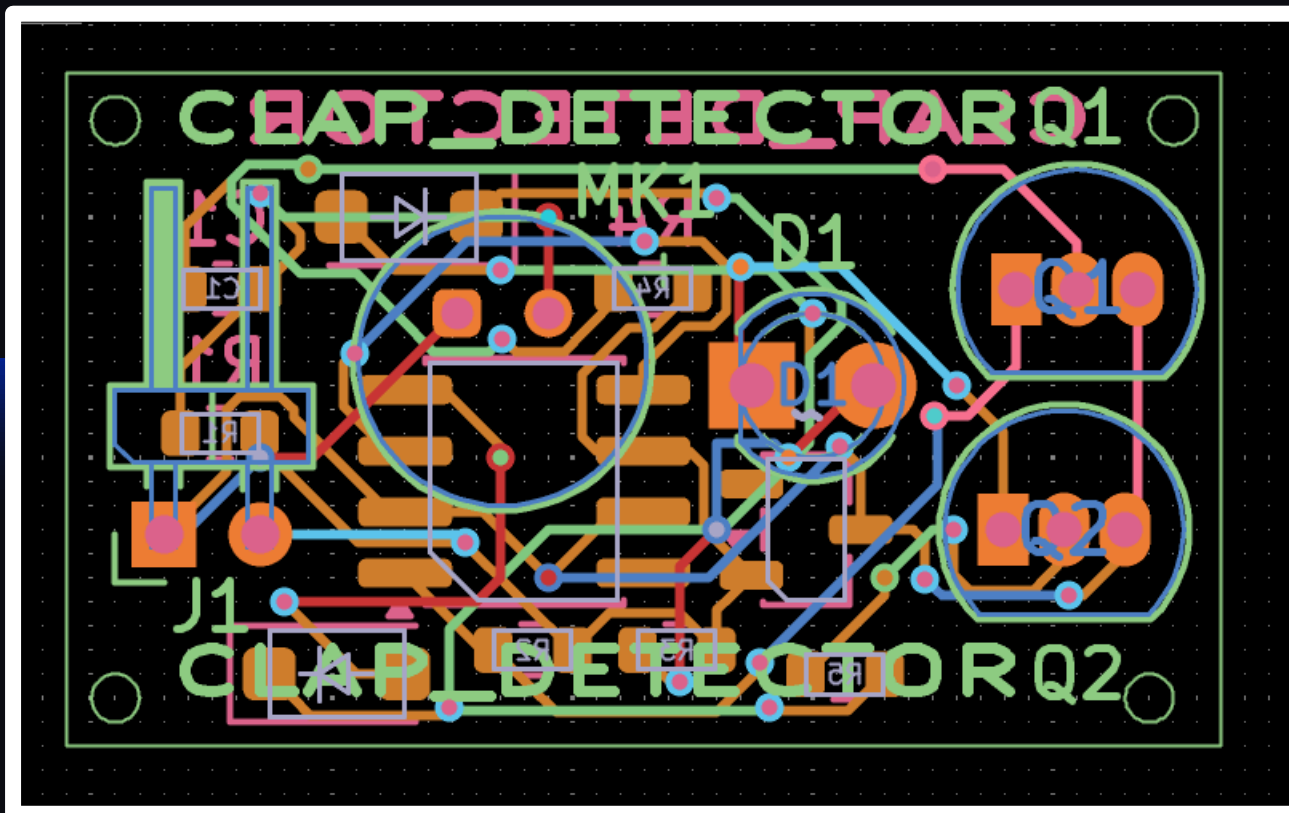
Drawing Sheet Editor

Edit drawing sheet borders and title blocks for use in schematics and PCB designs

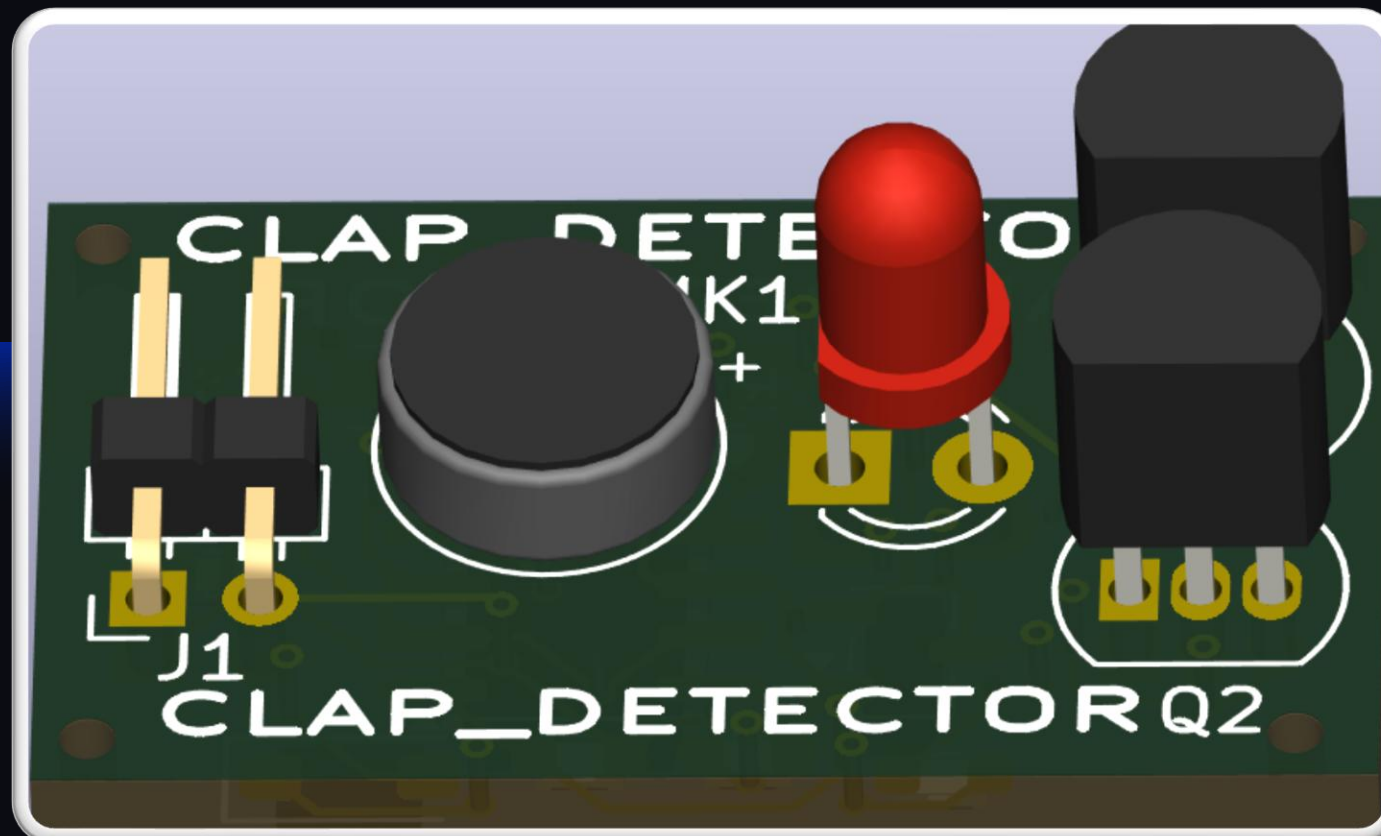


Plugin and Content Manager

Manage downloadable packages from KiCad and 3rd party repositories



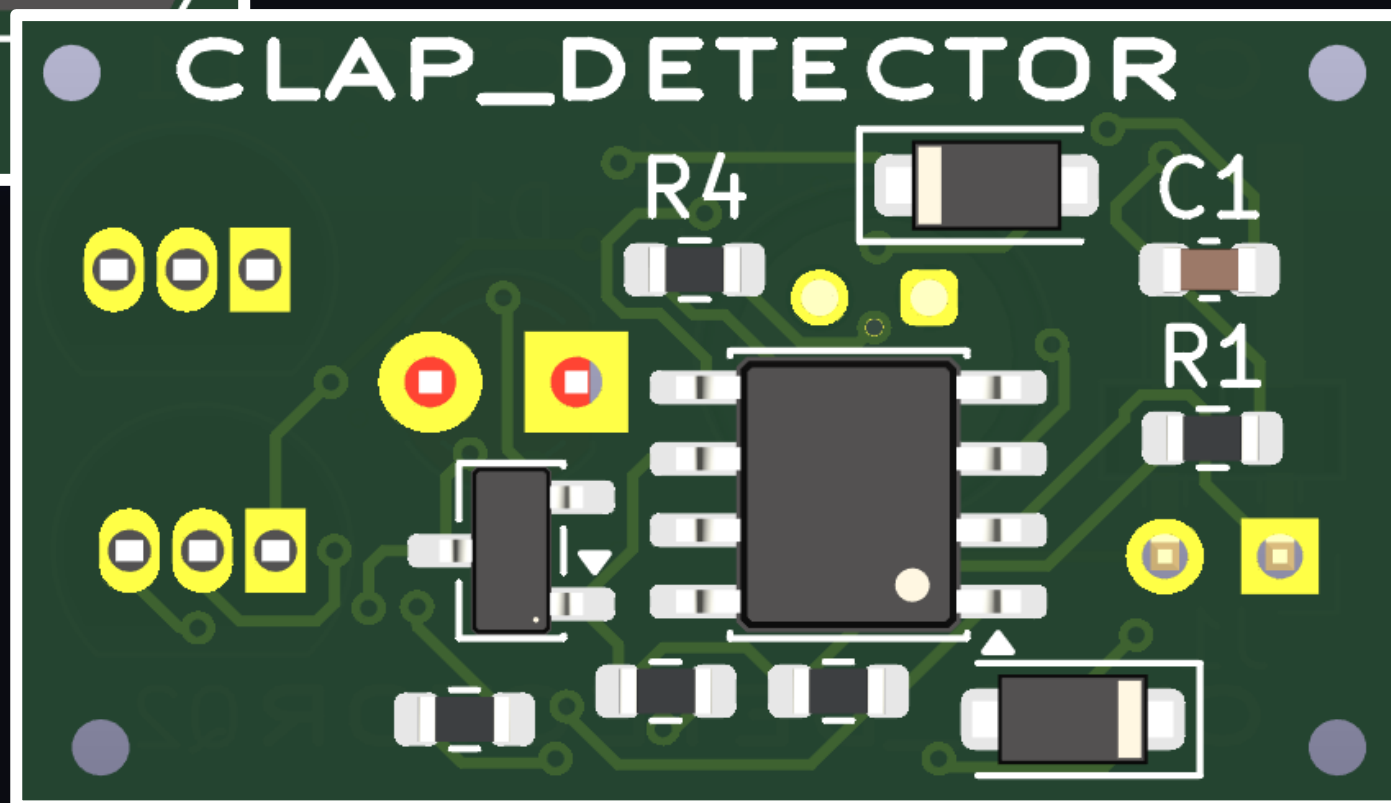
FINAL PCB STRUCTURE





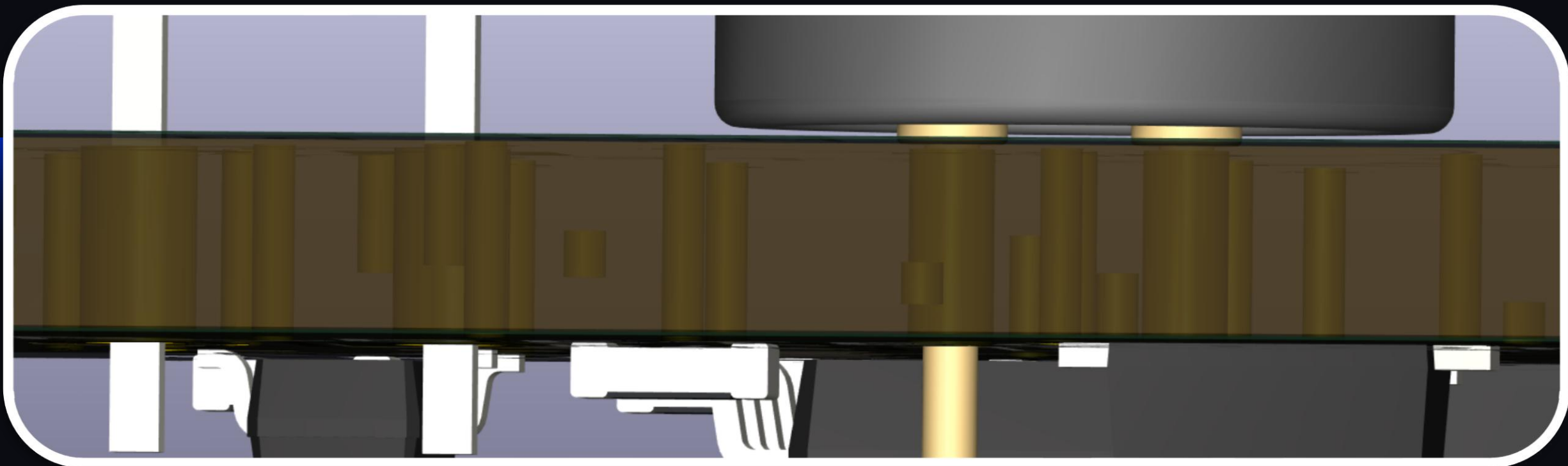
Z-axis VIEW

-Z-axis VIEW





Side View



APPLICATIONS

- **Hands-free light control** – Switch lights ON/OFF using a clap without touching the switch.
- **Home automation** – Can be used for touch-free control of household electrical devices.
- **Assistive devices** – Useful for people who have difficulty operating conventional switches.
- **Automatic presence indication** – The added proximity sensor detects a person approaching and provides an LED indication.
- **Touch-free control systems** – Can be used in rooms, labs, offices, or other areas where contactless operation is useful.

Advantages

- ✓ **Touch-free operation** – Devices can be controlled without physically touching a switch.
- ✓ **Simple and easy to use** – A clap provides a convenient method of switching.
- ✓ **Low-cost design** – Uses simple and easily available electronic components.
- ✓ **Improved convenience and accessibility** – Useful for users who may find conventional switches difficult to operate.
- ✓ **Additional proximity detection** – The added sensor detects a nearby person and provides a visual LED indication.

LIMITATIONS

- ❖ **False triggering** – Other loud sounds may be detected as a clap and activate the circuit.
- ❖ **Limited detection range** – The microphone and proximity sensor work only within their specified sensing ranges.
- ❖ **Environmental interference** – Background noise can affect reliable clap detection
- ❖ **Sensitivity variation** – Detection may vary depending on clap intensity, distance, and sensor placement.
- ❖ **Limited control** – The basic circuit is mainly suitable for simple ON/OFF operation and does not provide advanced control features.

FUTURE-SCOPE

- ❑ **Smart home integration** – Connect the circuit with IoT systems for remote and automatic appliance control.
- ❑ **Multiple appliance control** – Extend the circuit to control lights, fans, and other devices.
- ❑ **Improved clap recognition** – Use signal processing or AI to distinguish a clap from other sounds.
- ❑ **Advanced proximity sensing** – Add accurate distance measurement and automatic device activation.
- ❑ **Mobile/app control** – Integrate Bluetooth or Wi-Fi for controlling the system through a smartphone.

CONCLUSION

- The **clap switch** provides a simple and convenient method for **touch-free ON/OFF control**.
- The circuit successfully demonstrates **sound-based switching using a clap**.
- The added **proximity sensor** provides an additional feature for detecting a nearby person.
- The project uses **simple, low-cost, and easily available components**.
- Overall, the project demonstrates the practical application of **electronic switching, sound detection, and proximity sensing** in a single PCB.

REFERENCES

- ✓ **Clap switch circuit concepts** – how a microphone detects the clap and how the transistor stages perform switching.
- ✓ **Proximity sensor documentation** – working principle, pin configuration, and sensing operation of the sensor you used.
- ✓ **Component datasheets** – specifications and operating characteristics of components such as transistors, LEDs, diodes, resistors, and capacitors.
- ✓ **PCB design documentation** – schematic creation, component footprints, PCB layout, and routing.
- ✓ **Electronic circuit textbooks/web resources** – basic principles of transistors, switching circuits, microphones, and sensors

THANK YOU